

Polypropylene Plastic Waste Conversion to Lubricants over Ru/TiO₂ Catalysts

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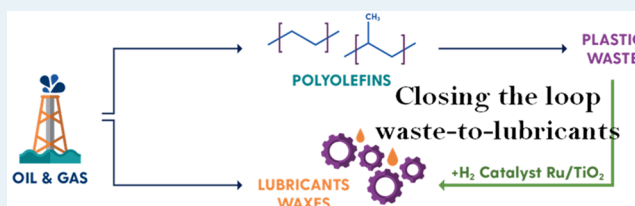
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ABSTRACT: Plastic recycling and upcycling are required to combat the environmental crisis from landfilling consumer products. Chemocatalytic technologies are the most promising approach to achieve this. Here, we show that ruthenium deposited on titania is an active and selective catalyst in polypropylene breakdown into valuable lubricant-range hydrocarbons with narrow molecular weight distribution and a low methane formation at low temperatures of 250 °C with a modest H₂ pressure.

Amorphous polypropylene and everyday bags and bottles were also effectively converted to lubricants with yields up to 80+%. Quantification of critical properties, including pour point, kinematic viscosity, and viscosity index, indicates that the products are promising alternatives to currently used base or synthetic oils. The reaction network involves the sequential conversion of polymer into the oil with a gradual decrease of molecular weight until ~700–800 g/mol and slow liquid gasification to methane and ethane. NMR, ATR-IR, GCMS, and isotopic labeling experiments expose the complexity of structure and reaction evolution whereby hydrogenolysis involves intermediate dehydrogenation with synchronous loss of polypropylene stereoregularity.

KEYWORDS: plastic waste, hydrogenolysis, lubricants, upcycling, circular economy



INTRODUCTION

Plastic production has experienced exponential growth in recent years, mainly because of the high demand for single-use packaging materials.^{1–3} The current pandemic restrictions spurred the need for food, pharmaceuticals, and cleaning packaging together with polymer-based protective equipment.^{4,5} In recent years, plastic waste has become a major threat to the environment because of the contamination of soil, oceans, and air.^{6–8} A vast amount of plastic waste accumulates in landfills due to it being unsuitable for mechanical recycling. Despite the significant success in the recycling of polyethylene terephthalate (PET),⁹ a negligible percentage of high- or low-density polyethylene (HDPE or LDPE), as well as polypropylene (PP), is recycled.^{2,3} For instance, one of the world's most used plastics—PP—has a global production rate of 55 × 10⁶ t/a with only 1% being recycled.^{10,11} Converting polyolefins to their respective monomers (ethylene or propylene) requires an enormous amount of energy due to unfavorable thermodynamics of C–C bond breaking making chemical recycling challenging.^{12,13}

Lately, the concept of upcycling polyolefin waste has been discussed extensively.^{2,10,14–17} Recent reports have shown that noble metal nanoparticles deposited on silica, alumina, ceria, or carbon in the presence of H₂ can easily break C–C bonds in the polyolefin backbone via hydrogenolysis producing low-molecular-weight wax or smaller fuel-range hydrocarbons.^{14,17,18} The mild reaction conditions, in this case, allow nearly 2-fold energy savings compared to plastic to fuel

conversion via pyrolysis.¹⁹ Ruthenium (Ru) is one of the most active D-metals for C–C bond hydrogenolysis of LDPE.^{17,18} One of the drawbacks of Ru-based catalysts and other metals,^{17,18} is its strong tendency to break terminal C–C bonds forming low-value methane and other light alkanes.^{17,18,20} While exciting progress is rapidly being made, evaluation of product properties has not been conducted yet. LDPE hydrogenolysis products are essentially linear alkanes, a property limiting their use as fuels or lubricants.^{14,15,17,18} The long reaction times, high temperatures, high gas production, and low polymer to catalyst mass ratios, evidenced in recent reports, expose some of the challenges in polyolefin conversion and the need to discover more active and selective catalysts.

We propose that controlled hydrogenolysis could pave the way toward the commercially attractive conversion of plastic waste to lubricants (Figure 1). Lubricants represent a large >37 × 10⁶ t/a market²¹ driven by automotive engine oils, industrial metalworking fluids, and hydraulic oils.²² Nearly half of the lubricants are mineral base oils (C₂₀–C₆₀ isoalkanes),^{23,24} produced by mild hydrodewaxing of crude oil distillates.²⁵ These molecules could be made via the hydrogenolysis of

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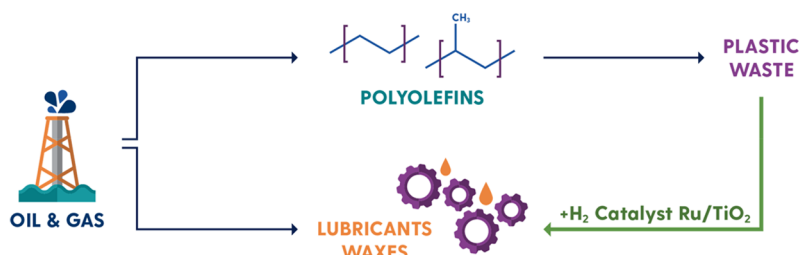


Figure 1. Proposed upcycling of plastic waste via hydrogenolysis to enable a circular economy.

Table 1. Catalyst Screening for i-PP Hydrogenolysis^a

catalyst	metal loading according to XRF (wt %)	yields (%)			DSC analysis of solid		GPC analysis of liquid		
		gas (C ₁ –C ₆)	liquid	solid	T _m (°C) ^b	fwhm of melting peak (°C) ^b	M _w (kDa)	M _n , kDa	PDI
transition metals on TiO ₂ support									
Pd/TiO ₂	5.6	0.6	0.0	98.7	171	15.3			
Rh/TiO ₂	5.1	1.4	4.9	92.8	161	8.0			
Ir/TiO ₂	5.0	0.4	0.0	93.5	166	10.4			
Ni/TiO ₂		0.1	0.0	≥99	168	14.0			
Pt/TiO ₂	4.5	0.3	0.0	≥99	166	13.5			
Ru/TiO ₂	5.9	28.2	65.6	6.4	113	70.9	0.90	0.68	1.3
Ru on different supports									
Ru/SiO ₂	4.9	65.3	23.6	5.7			0.66	0.61	1.1
Ru/C	5.0 ^d	100.0 ^e	0.0	0.0					
Ru/Al ₂ O ₃ ^e	5.1	15.8	44.7	28.5			1.36	0.96	1.4
Ru/CeO ₂	4.8	97.8	6.8	0.0			0.33	0.32	1.0

^aReaction conditions: 250 °C, 16 h, 30 bar H₂ pressure, 2 g i-PP loading, 0.1 g of catalyst. ^bFor raw i-PP: T_m equals 168 °C with fwhm of 10.0 °C. ^c8 h reaction time. ^daccording to the manufacturer. ^equantitative conversion to gas within experimental error. PDI stands for polydispersity index.

polyolefins. Given the large market and the increasing demand for lubricants and their higher value than fuels and most plastics, this can be a low-hanging fruit upcycling strategy. The challenge is to avoid overcracking of polyolefins to gases by mild C–C bond breaking and encoding branching into the end-product backbone to make it valuable.

Here, we study the hydrogenolysis of previously unexplored PP plastic waste over a Ru/TiO₂ catalyst and demonstrate the catalyst's unique properties. The abundance of CH₃ groups on the carbon backbone may allow the production of highly branched, potentially viable synthetic oil components. We measure lubricant properties to assess and showcase the potential of such products for upcycling. Mechanistic studies via isotopic labeling and spectroscopy bring intriguing insights into the reaction network and polymer activation.

RESULTS AND DISCUSSION

Catalyst Screening: Effect of Metal and Support.

Isotactic polypropylene (i-PP) with a moderate molecular weight was selected as the primary substrate. Several transition metal (M) nanoparticles were deposited on TiO₂ and screened at 250 °C (Tables 1, S1, S2), a relatively low temperature where chemistry happens in the polymer melt. The catalyst amount is lower than most prior works.^{17,18} Most of the M/TiO₂ catalysts show nearly no conversion of the i-PP. Experiments without a catalyst and with pure TiO₂ support showed no detectable formation of liquid with the polymer melting point remaining unchanged (Table S2). In case of M/TiO₂ the remaining solid also possesses a melting point of ~168 °C, similar to the initial polymer. The increased peak width, evidenced by differential scanning calorimetry (DSC), corresponds to minor catalyst–polymer interactions.

In the case of *n*-octadecane hydrogenolysis, Pt shows no activity at 250 °C.¹⁷ LDPE/HDPE hydrogenolysis over a Pt-based catalyst requires temperatures >300 °C or longer reaction times^{14,15} than used here, consistent with our experiments' low Pt/TiO₂ activity for i-PP. The same holds for Ni/TiO₂ that requires a higher reaction temperature for hydrogenolysis. Rh/TiO₂ exhibits a low activity toward the formation of liquid and reduces the melting temperature of the solid residue by 7 °C. Previous studies showed that Rh/C is moderately active in hydrogenolysis of *n*-octadecane¹⁷ and smaller alkanes.²⁶ Overall different metals supported on TiO₂ under low-temperature conditions show an activity trend very similar to polyethylene hydrogenolysis.^{17,18}

Ru/TiO₂ catalyst forms significant fractions of liquid oil (~66%), light C₁–C₂ hydrocarbons (~25%), and a minor fraction of C₃–C₆ gases (~3–4%). The outstanding performance of Ru/TiO₂ correlates with its well-known activity in small alkanes hydrogenolysis. The Ru-based catalyst has been found to possess the highest activity for ethane, *n*-heptane, and neopentane hydrogenolysis compared to all other transition metals.²⁷ To evaluate the role of the support, Ru was deposited on several carriers (Table 1). All Ru-based catalysts, except Ru/Al₂O₃, are highly active, consistent with the Ru's remarkable hydrogenolysis activity on light alkanes, squalene, and PE at 200–250 °C documented before.^{17,18,20} However, all form major fractions of light C_{1–4} hydrocarbons with up to 82% yield to methane (Table S1 and Figure S1) except for Ru/TiO₂. The methane formation decreases in the order Ru/C > Ru/CeO₂ > Ru/SiO₂ > Ru/Al₂O₃ ≈ Ru/TiO₂. The liquid yield over Ru/TiO₂ is the highest. The high activity of Ru/C and Ru/CeO₂ to light gas is consistent with a recent report on LDPE conversion.^{17,18} The structure of Ru/TiO₂ analyzed by

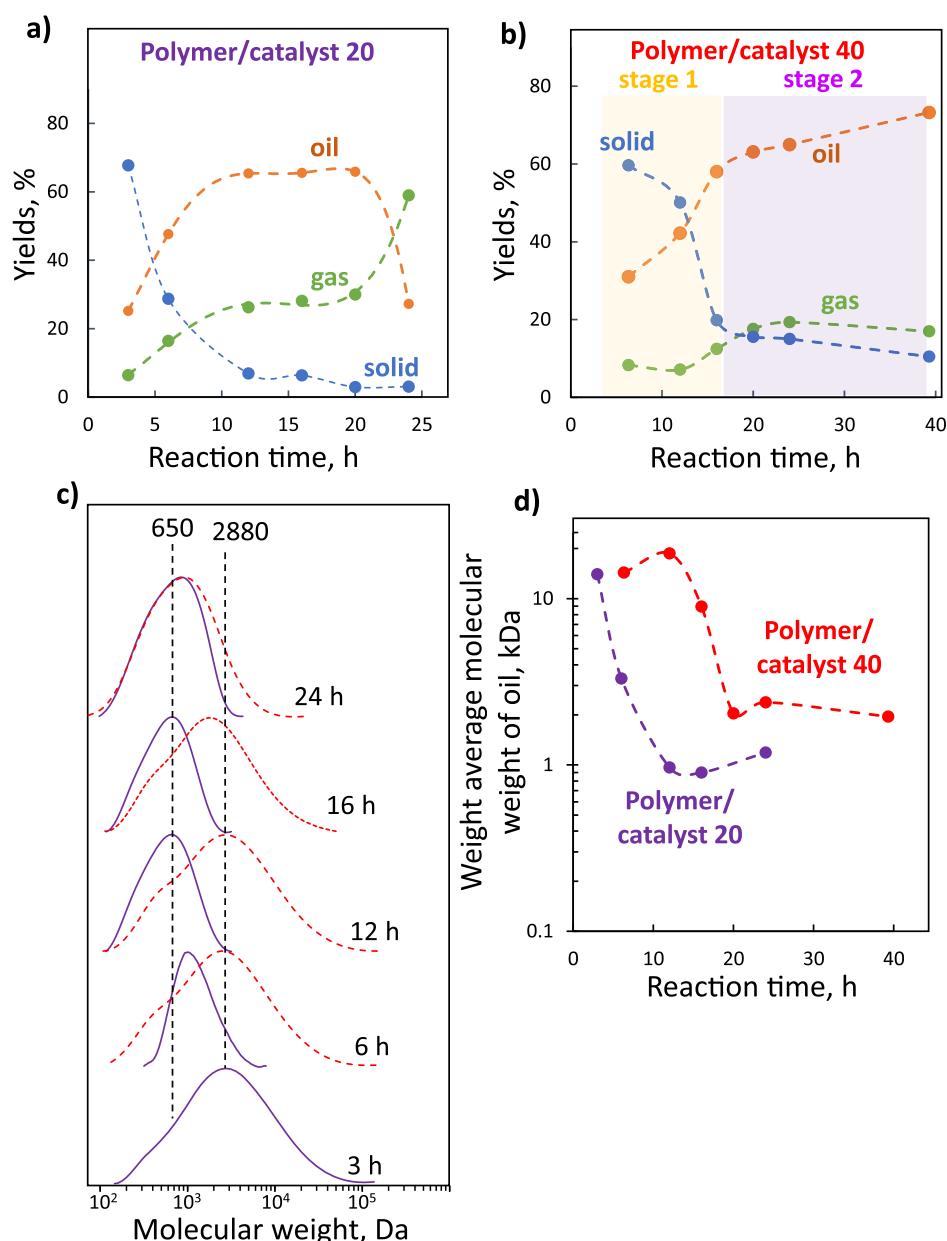


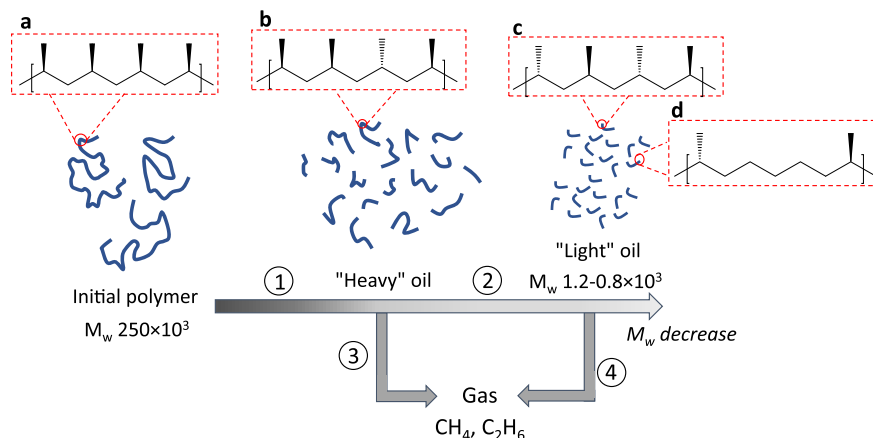
Figure 2. Time-dependent isotactic polypropylene (i-PP) hydrogenolysis over Ru/TiO₂. Yields vs reaction time of major products with polymer/catalyst weight ratio of 20 (a) and 40 (b) over Ru/TiO₂. (c) Molecular weight distributions of liquids at different reaction times derived from GPC for polymer/catalyst ratio 20 (solid purple lines) and 40 (dotted red lines). (d) Effect of polymer/catalyst ratio on M_w vs reaction time. Reaction conditions: 250 °C, 30 bar H₂, polymer loading 2 g, catalyst loadings 0.1 or 0.05 g (polymer catalyst ratio 20 and 40, respectively). Stages 1 and 2 are indicated as shaded areas in panel b based on the molecular weight distribution of panel d.

XRD, STEM, and H₂ chemisorption (Figure S2 and Table S3) indicates small 1.2 nm Ru nanoparticles on the surface of TiO₂ anatase nanoparticles. According to recent results,¹⁸ several transition metal oxides stabilize small Ru particles of <1.5 nm, while Ru/C has larger particles of ~2.6 nm. It is known that small alkanes hydrogenolysis, especially of extensively branched substrates, over Ru nanoparticles is a highly structure-sensitive reaction.²⁸ Thus the differences of Ru-based catalysts in terms of methane selectivity may be caused by differences in particle size. This topic deserves further work.

The oil molecular weight distribution was determined by gel permeation chromatography (GPC). For all Ru-catalysts, both the number-averaged (M_n) and weight-averaged (M_w) molecular weights of the liquid are below 1.5 kDa, significantly

lower than the initial i-PP substrate ($M_w \sim 250$ kDa). Ru/CeO₂ produces an oil with the lowest M_w and the highest C₁–C₄ gas yield (Table 1).

Nakaji et al.¹⁸ showed that LDPE hydrogenolysis over 5% Ru/CeO₂ produces liquid hydrocarbons of low molecular weight at a longer reaction time (72 h) than used here; their results are qualitatively consistent with ours despite using different feedstocks. Ru on various supports produced mainly linear, low molecular weight C₅–C₂₁ range hydrocarbons (~40–55 wt %) with some light gas and wax (a total up to 33%).^{17,18} Taken together, we notice a remarkable effect of the plastic's architecture on the product; LDPE converts mainly to lighter hydrocarbons, whereas PP forms a heavier oil. Our results also expose that the combination of the metal and the

Scheme 1. Reaction Network of Successive Polypropylene Hydrogenolysis over Ru/TiO₂^a

^aThe structure of the liquid products: (a) initial polymer, ideal i-PP sequence, (b) heavy oil, distortion of stereoregularity, (c) light oil, further distortion, and (d) light oil, loss of CH₃ groups due to late-stage demethylation.

support is crucial and dictates the activity and selectivity to light gases and valuable liquids. We elaborate on these exciting effects of microstructure and catalyst below. Since Ru/TiO₂ is the most active and selective catalyst in converting i-PP to heavy oil, it was selected for further studies.

Effect of Reaction Time on Product Distribution and Depolymerization Reaction Network. Reaction profiles for a polymer to catalyst weight ratio of 20 are shown in Figure 2a. The oil yield increases up to 66% with a concomitant decay of the residual solid below 10% and the formation of mainly C_{1–4} hydrocarbons (consisting of 85% methane) with a total gas yield up to 28% after 16 h. Detailed information on the product distribution in the gas and liquid is reported in Tables S4 and S5. Analysis indicates a rapid decrease of the oil M_w from an initial value for i-PP of 250 kDa to 14.0 and 0.97 kDa after 3 and 12 h of reaction, respectively (Figure 2c). The polydispersity index (PDI), which reflects the broadening of the molecular weight dispersion (Table S5), also decreases from 2.7 to 1.3–1.4 with time due to the sequential consumption of the heavy portion of the distribution and the formation of relatively light oil. Upon a rather rapid change (stage 1) in the first ~10 h, the liquid products exhibit constant M_w and PDI in the 12–24 h time interval (stage 2) because of the slow hydrogenolysis. Notably, a further increase of the reaction time to 24 h leads to excess gas formation at the liquid's expense. Interestingly, no intermediate species with a molecular weight between light oil and gas form at any reaction time. The oil produced in 12–24 h time intervals is denoted as light. Heavy oil refers to liquid products with a molecular weight higher than light oil.

An approximate turnover frequency for the Ru/TiO₂ catalyst is estimated using M_n of initial i-PP and light oil formed after 16 h. Calculations indicate an average of 3.2 broken C–C bonds per Ru surface site per second, which lays in the same range as the most active Ru/TiO₂ catalysts in small alkane hydrogenolysis.²⁹

Experiments with a polymer/catalyst weight ratio of 40 (Figure 2b–d) show that reduced catalyst loading mainly affects the solid-to-heavy oil conversion rate. Specifically, the oil yield increases linearly with reaction time toward the previously observed value of ~60% with a significant solid fraction (16–20%) left even after 20 h of reaction time. The oil fraction has a substantially larger M_w at short reaction times

(Figure 2c, d), and it takes ~20 h to achieve a molecular weight of ~2.0 kDa. We observe a constant M_w oil up to 40 h reaction time during which 90% of the solid is consumed. A lower catalyst loading slows down both the heavy oil hydrogenolysis to light oil and its parallel gasification leading ultimately to improved selectivity to liquid products with less gas—the maximum yield of gas is just 16% after 20 h for the polymer/catalyst ratio of 40 compared to 28% after 16 h for the polymer/catalyst ratio of 20, discussed above. The unexpected low methane production over Ru/TiO₂ at short reaction times (a catalyst known to produce chiefly methane) at low catalyst loadings departs profoundly from all prior reports, typically invoking high catalyst amounts.

Transient profiles (Figure 2a, c) show the light oil formed after 12 h is reactive for hydrogenolysis and secondary gas formation. On the basis of the molecular weight distributions and gas formation curve (Figure 2a,c), heavy oil contributes more to gas formation than light one at low reaction time. Comparison of curves in Figure 2a and b reveals that gas is a secondary product, produced from the heavy oil and not from the initial polymer. The plastic conversion is sequential, forming heavy oil (stage 1) that further reacts to light oil (stage 2). Lower catalyst loadings promote the heavier oil product and enable tuning the product molecular weight distribution. This preference of the catalyst for longer chains is intriguing and is discussed further below.

Previously, a processive mechanism was proposed for HDPE hydrogenolysis over mSiO₂/Pt/SiO₂ catalysts,¹⁵ where a polymer chain undergoes complete deconstruction to small pieces with no intermediates desorbing from the surface until sufficiently cracked. The authors¹⁵ observed a nearly constant molecular weight distribution of the residual polymer over a broad range of conversions with a constant bell-shaped distribution of low molecular weight hydrogenolysis products. This insensitivity of product distribution to conversion was attributed to the specific porous structure of silica, which enables only polymer adsorption with no release until complete deconstruction. The continually shifting distribution of molecular weight (Figure 2c) indicates a different reaction mechanism for PP over Ru/TiO₂. Hydrogenolysis occurs through multiple adsorption–desorption events, starting with relatively large chains gradually converted to lighter products that desorb and may re-adsorb. The results obtained with

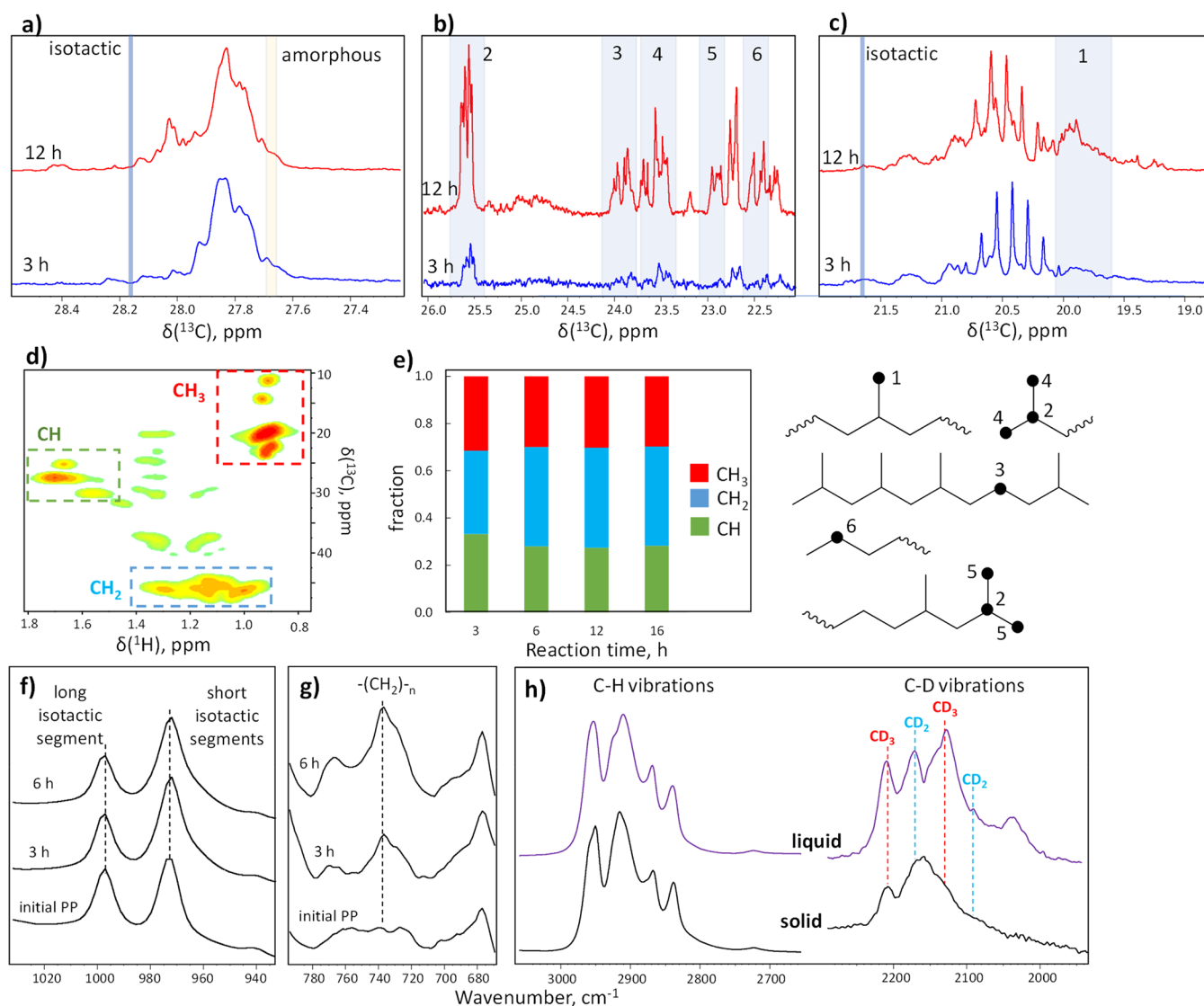


Figure 3. Liquid product characterization using NMR and ATR-FTIR. ^{13}C NMR spectra liquid products after 3 and 12 h reaction in (a) CH groups resonance region, (b) in the 26–22 ppm region, and (c) in CH_3 groups resonance region. ^{13}C – ^1H HSQC NMR of liquid after 12 h reaction (d) and distribution of different CH_x groups calculated from NMR (e). ATR-IR spectra of solid residue (f, g). ATR-IR spectra of solid and liquid products after 6 h reaction of i-PP with D_2 (h). Reaction conditions: 250 $^\circ\text{C}$, 30 bar H_2 , polymer loading 2 g, polymer/catalyst weight ratio of 20.

different polymer/catalyst ratios point to a plausible, simplified reaction network (Scheme 1). The initial polymer continually undergoes C–C bond hydrogenolysis producing mainly a heavy oil (stage 1 in Scheme 1). In the course, the internal C–C bonds of the heavy oil crack to light oil (stage 2 in Scheme 1). Both oils are gasified to light hydrocarbons, mainly methane and ethane (stages 3 and 4 in Scheme 1). At low reaction time, gas is produced mainly from heavy oil. At longer reaction time and high catalysts concentration, significant conversion of light oil to gas also occurs. The Ru surface has a lower affinity toward holding on reaction intermediates allowing their release back into the melt after C–C bond breaking, similar to LDPE hydrogenolysis over Ru/C and Ru/CeO₂.^{17,18} Given the diversity in supports, the release of intermediates into the melt and the associated mechanistic differences among catalysts appear to be associated mainly with the metal itself. The desorption of products is critical to producing higher molecular weight liquids.

The hydrogen gas solubility in the melted polymer at reaction conditions is expected to be >10 mg/g;³⁰ yet, it is

hard to estimate to what extent PP and hydrogen compete for catalyst active sites. Generally, H_2 is expected to coadsorb with the polymer.

Structure of Oil Products and Mechanistic Implications. Liquid samples produced over Ru/TiO₂ with a polymer/catalyst ratio of 20 were selected for a detailed structural study. NMR spectroscopy has been employed for characterizing the polypropylene microstructure,³¹ some of its oligomers,^{32–35} and the lubricant oil.³⁶ ^1H NMR spectra of liquid products consist of three regions associated with the CH, CH_2 , and CH_3 groups (Figure S3).³⁷ The liquid product spectra deviate significantly from the initial i-PP, indicating that the oil is not merely a short-chain oligomer of i-PP. ^{13}C NMR with proton decoupling and 2D ^{13}C – ^1H HSQC NMR (Figures 3a–d and S4–S8), which are sensitive to the monomer sequence and the overall structure of long paraffins,³⁸ were employed. Three main functional groups were observed in both spectra, similar to the initial i-PP, consistent with the ^1H NMR results (Figure S3). In the CH resonance region, after 3 h of reaction, most peaks shift from

28.2 ppm (typical for CH in the ideal i-PP chain³⁹) to lower values. The PP chain conformation deviates from the perfect isotactic sequence with more randomly oriented CH₃ groups leading to peak shifting upfield, typical for amorphous PP.^{40,41} The intermediate peaks centered at ~27.9 ppm reflect the different sequences of nonequivalent steric PP pentads with some combination of meso and racemo configurations (Scheme S1). Scheme 1 depicts this transition from isotactic structure (a) to a more randomly oriented structure (b) in lighter oil. The isotactic sequence distortion observed already after 3 h indicates that the liquid product possesses substantial configurational reorientation of CH₃ groups in addition to C–C bond breaking.

In the CH₃ region, a similar deviation from the standard i-PP spectra is clearly seen³⁸ (Figures 3c and S5). The CH₃ group signal at 21.75 ppm in i-PP disappears, giving rise to at least 6–7 new resonances, typical for stereospecific pentads with more random orientation of CH₃ groups.⁴² Beyond the elimination of isotacticity, the data shows a broad signal at ~20.0 ppm at long reaction times because of CH₃ group attached to the (CH₂)_n backbone (Figure 3).⁴³ The NMR results indicate the stereoregular CH₃ orientation in the PP is completely disrupted in the oil. Apparently, the formation of oil by its very nature involves significant changes in the polymer configuration. The conformational change in PP is characterized by a rotational barrier of ~12.5–17 kJ/mol,⁴⁴ an order of magnitude lower than the C–C bond-breaking activation energy in alkane hydrogenolysis.^{45,46}

Figure 3c shows additional ¹³C resonances because of demethylated fragments of the liquid products, untypical of stereoregular PP. The resonances (3 in Figure 3c) at ~24.0 ppm correspond to a (CH₂)₃ sequence in the middle of regular PP due to the breaking of one CH₃–CH bond in the parent structure.⁴⁷ Peaks at 23.0 ppm (2 and 5 in Figure 3c) arise because of isobutyl groups at the end of a mixed polyethylene-polypropylene chain.⁴⁸ The bands due to the PP chain end groups³¹ (2 and 4 in Figure 3c) increase in intensity over time in line with the molecular weight decline (Figure S7). Ethyl groups at chain-ends give rise to peaks at ~22.5 ppm (6 in Figure 3).^{49–51} Because the partially demethylated polymer, other resonances are seen in the 40–30 ppm range in Figure S8. Quantitative analysis shows that the highly branched oil formed in the first 3–6 h is less stable with a tendency for =CH–CH₃ bond breaking at long reaction times. Model studies with branched alkane conversion on metal surfaces reveal that branching indeed enhances their dissociation probability.⁵² So we argue that the lower stability of the heavy oil is partially caused by the higher content of CH₃ groups on the carbon backbone. Once they are removed by demethylation oil becomes more stable toward further hydrogenolysis.

The ¹H and ¹³C NMR data allows quantification of the CH_x functional group fractions vs time (Figure 3e). This analysis does not include peaks of chain ends and demethylated fragments. After 3 h of reaction, the liquid exhibits a very similar distribution to the initial i-PP, with a 0.33 fraction for each group. The methine content drops by 15–20% at longer reaction times because of the concurrent increase in CH₂ with constant CH₃. These results show that part of the oil has a similar CH_x distribution to PP, and another has demethylated fragments reminiscent of an ethylene–propylene copolymer. The molecular weight distributions (Figure 2) and NMR results combined provide the following insight into the PP conversion (Scheme 1). Initial PP hydrogenolysis (stage 1 on

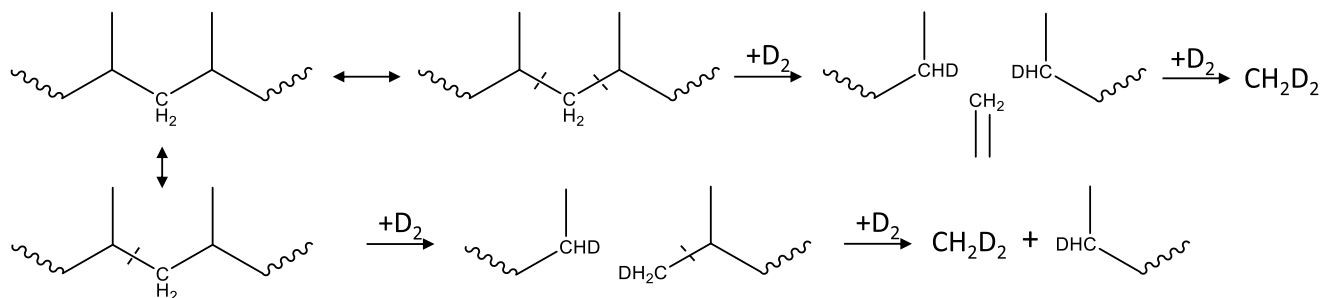
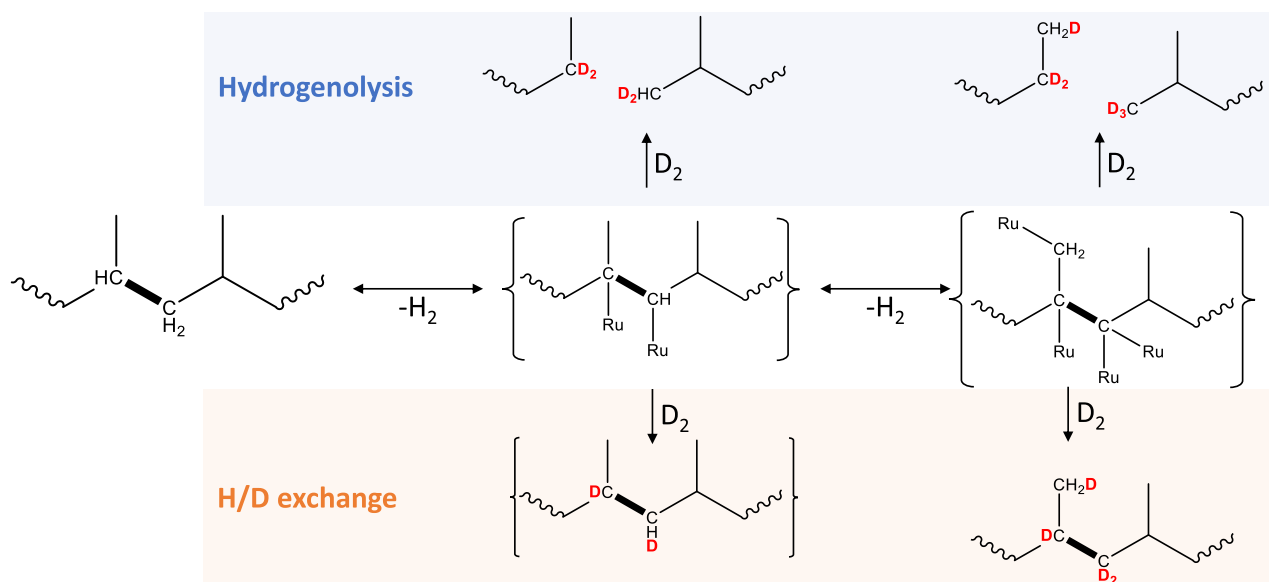
Scheme 1) exhibits a reduction in the molecular weight with loss of stereoregularity (structures a and b) in Scheme 1). At longer reaction times, secondary hydrogenolysis (stage 2) leads to light oil with short PP chains (structure c) and partially demethylated short chains with (CH₂)_n segments (structure d).

Attenuated total reflectance-infrared (ATR-IR) spectroscopy analysis of the solid residue shows a band at 998 cm^{−1} due to 10–11 isotactic PP units in a segment forming a 3₁ helical sequence⁵³ and a band at 973 cm^{−1} because of the shorter isotactic helices with 2–4 monomers units (Figures 3f and S9).⁵⁴ The I₉₉₈/I₉₇₃ ratio decreases by 40% in the first 3 h compared to the initial i-PP indicating loss of isotacticity in the solid residue (Figure S10). Interestingly, the isotacticity remains constant between 3 and 6 h of reaction where significant consumption of the raw polymer occurs (Figure S10). Apparently, the initial rapid changes in the solid cannot wholly eliminate isotacticity and randomly orient all CH₃ groups, as it happens in the liquid part of the products. Similar to the oil, a minor amount of relatively long (CH₂)_{n>5} sequences appears at ~740 cm^{−1} because of partial demethylation.⁵⁵ No significant differences are visible in the C–H stretching region with reaction time (Figure S9), indicating a relatively similar content of various CH_x groups in all solid residues. With a lower catalyst loading (Figure 2c), the solid residue retains ~70% of the initial isotacticity after 6 h (Figure S9). Since the solid reacts slower in this case, isotacticity decays relatively slowly up to 20 h. On the basis of this finding, we hypothesize that the loss of stereoregularity is a prerequisite of C–C bond hydrogenolysis in the early stages of the reaction when the M_w is still high.

Alkane hydrogenolysis necessitates intermediate dehydrogenation of the reactant.^{18,56} This pathway was analyzed for PP depolymerization by replacing H₂ gas with D₂ followed by ATR-IR, GCMS, ¹H NMR, and ²H NMR monitoring of reaction products. Comparison of the NMR spectra of liquid products obtained with H₂ and D₂ as reactants showed that all CH_x groups are evenly deuterated with 20% intensity loss in the ¹H spectra (Figure S11 and Table S6). ATR-IR revealed the appearance of new C–D stretching (Figure 3h) and δ(CD_x) bending (Figure S12) vibrations in the deuterated liquid. The intensity of the CD₃, CD₂, and CD-related bands resembles the original C–H intensity distribution (Figure 3h). The H₂/D₂ isotope exchange leads to nearly no effect on product yields (Figure S13). This indicates that D₂ does not affect the initial consumption of the i-PP or the gas production early in the reaction (stages 1 and 3 in Scheme 1). GPC analysis revealed a substantial increase in both the M_w and PDI of the liquid produced in D₂ (Figure S13 and Table S6). Thus, isotope exchange slows down the light oil production (stage 2 in Scheme 1), implicating hydrogenation/dehydrogenation steps as kinetically influential reactions.

According to GCMS analysis (Figure S14), CH₂D₂ and C₂H₄D₂ are the preferred isotopomers of methane and ethane, respectively. Other gaseous products (C₃–C₅) reveal a statistical distribution of deuteration. The probability to get *N* deuterons in a molecule is fitted by a Poisson distribution.⁵⁷ The average numbers of deuterons were 2.9, 2.4, and 4.5 for propane, butane, and pentane, respectively. So, C₃–C₅ gas hydrocarbons and liquid products are deuterated statistically according to the standard hydrogenolysis mechanism.^{18,56} In the case of LDPE hydrogenolysis over Ru/CeO₂, for example, before C–C bond breaking, the polymer undergoes extensive

Scheme 2. Concerted and Sequential Hydrogenolysis Pathways to Methane

Scheme 3. Two Extreme Pathways of D Insertion in Polypropylene CH₃ Group: C–C Bond Hydrogenolysis and Standard H/D Exchange

dehydrogenation on the metal through quasi-equilibrated C–H bond-breaking steps.^{18,58} This fact determines the high content of deuterons in the products since they form from deeply dehydrogenated intermediates.

The high probability of 2 deuterons in methane and ethane with a nonrandom distribution reflects that dehydrogenation of C₁–C₂ surface species may not be fully equilibrated. In this case, two D atoms are incorporated in methane and ethane after C–C bond breaking. Considering the structure of the liquid found from NMR (Figure 2e), we propose possible pathways for methane formation given the low content of CH groups and the higher content of CH₂ with the isotope composition of methane (Scheme 2).⁵⁸

Scheme 2 implies the conversion of two CH groups into two CH₂ groups without the formation of CH₃ groups. The surface-bonded methylene is rapidly hydrogenated to deuterated methane.⁵⁹ Surface mediated alkyl and alkylidene species also have a probability of exchanging hydrogen, leading to deviation of the number of deuterons from 2, as expected from Scheme 2. Another hypothesis proposed for the Ru-catalyzed hydrogenolysis of squalane⁵⁸ entails the C–C bond hydrogenolysis in tandem. This sequential pathway also leads to the observed isotope composition of gas and the structure of the liquid (Scheme 2).

The ATR-IR spectra of the solid residue are remarkably different from the liquid and gas products (Figure 3h). The

intensity of CD₃ stretching and bending vibrations is generally lower than in the liquid. The C–H stretching region in both the oil and solid due to unexchanged groups, on the contrary, is very similar. The distinction in the C–D stretching region is attributed to the less reactive CH₃ groups in the solid toward H/D exchange. Deuteration of methyl groups in the solid PP can be kinetically limited due to a high dehydrogenation energy barrier of the CH₃ group. The formation of the CD₃ group may be directly linked to hydrogenolysis, according to Scheme 3. So, the liquid will inevitably have more deuterons in methyl groups than the solid due to hydrogenolysis (Scheme 3).

The i-PP hydrogenolysis initiates with dehydrogenation, similar to small alkanes and LDPE.^{17,18,45} Unlike other substrates, the PP chain in contact with the surface should involve dehydrogenation of the CH₃ group to break C–C bonds in the polymer backbone. This explains the complete loss of isotacticity of the oil and partial loss of isotacticity in the solid residue. Indeed, dehydrogenation of the CH₃ group (Scheme 3) should lead to loss of stereoregularity of the CH₃ group orientation.

Effect of Polymer Feedstock and Lubricant Properties. A challenge in plastic chemical upcycling is the variability of the waste feedstock. It is essential to assess whether the Ru/TiO₂ catalyst is agnostic to different molecular weight distributions and stereoregularities. Diverse feedstocks provide

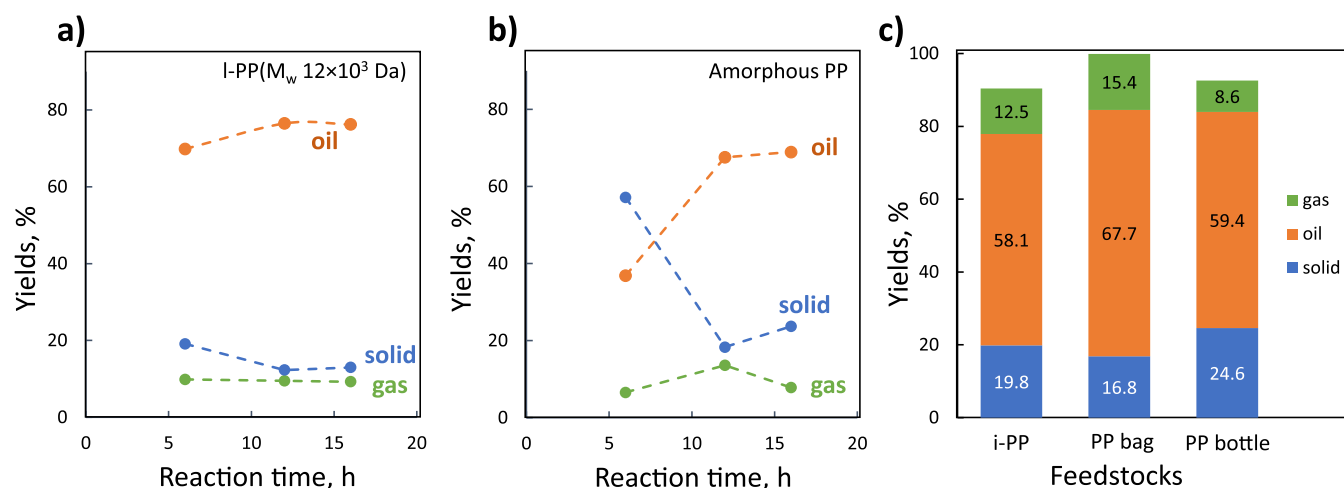


Figure 4. Effect of feedstock on PP hydrogenolysis. Yields of major products vs reaction time using low molecular weight i-PP (a) and amorphous PP (b). (c) Yields for a PP bottle (20 h reaction time) and bag (16 h reaction time). Reaction conditions: 250 °C, 30 bar H₂, catalyst Ru/TiO₂ 50 mg, substrate 2 g.

Table 2. Key Lubricant Properties of Oil Produced from i-PP over Ru/TiO₂ and Reference Samples

sample	KV40 (cSt) ^a	KV100 (cSt) ^a	VI ^b	pour point (°C)	DSC oxidation onset temperature (°C)
light lubricant	30.3 (±0.2)	6.2 (±0.1)	146 (±7)	−15	179
heavy lubricant	113.5 (±0.8)	42.2 (±0.3)	149 (±7)	−18	175
Chevron PAO6 ^c	30.8	5.9	139	−62	246
Canon no. 35 standard ^f	32.8 (±0.2)	5.7 (±0.1)	114 (±5)	−15	209

^aKV100 and KV40 are kinematic viscosities at 100 and 40 °C, respectively. ^bVI: Viscosity index calculated from KV100 and KV40. ^cCommercial PAO6 synthetic base oil manufactured by Chevron. ^fViscosity standard. The properties of commercial products were obtained from the specifications datasheet disclosed by the manufacturers. Reaction conditions: 250 °C, 16 h, 30 bar H₂, 2 g of i-PP. Light lubricant: 100 mg Ru/TiO₂; heavy lubricant: 50 mg Ru/TiO₂.

additional confirmation about the proposed reaction scheme for i-PP.

For a low molecular weight i-PP (i-PP, $M_w \sim 12\,000$ Da), the solid drops to ca. 15% in 3 h with a high liquid yield above 80%. At longer reaction times, the overall mass yields are almost stable with minor production of gas. In the same time scale, the M_w and M_n of the oil decrease to values characteristic of i-PP (Table S7). Because of the low molecular weight of i-PP, the heavy to light oil conversion (stage 1 in Scheme 1) is faster. So, the reaction entails the direct conversion of the melt to light oil (stage 2 in Scheme 1). In amorphous PP (a-PP), we observe a steep decrease of solid residue yield after 12 h with a concomitant oil yield increase to 65–70%. This transient profile looks similar to i-PP conversion with a shift to shorter reaction times (Figure 2d). The M_n of a-PP liquid products is nearly 1.3 times higher than i-PP after 12 h. This result indicates a considerable impact of polymer configuration and microstructure on the hydrogenolysis' early stages. PP-bags were effectively converted to oil with a yield above 70% (Figure 4c). Polystyrene additive can cause a somewhat lower oil yield for the PP-bottle. The product molecular weight distribution from all plastics resembles that of the model i-PP, that is, the liquid products are practically independent of the initial polymer structure. Upon regeneration by the reduction in hydrogen, the catalyst activity is recovered, giving the comparable oil yield produced by a fresh catalyst (Figure S15).

Lubricant properties are important in the evaluation of the real-world potential of the proposed polymer deconstruction strategy. Table 2 compares the lubricant properties of plastic

conversion products quantitatively to standard commercial products.

A higher viscosity index (VI) reflects a smaller dependence of lubricant's viscosity on temperature. Generally, an oil with a higher VI is desirable due to having a stable lubricating film over a wide temperature range. The VI is below 120 for Group I–III mineral oil^{60,61} and higher for synthetic lubricants.⁶² The plastics-derived base oils show an overall higher viscosity than polyolefin (PAO) commercial analog, but a higher VI. The pour point of base oils is higher than that of the commercial PAO. We hypothesize that the high pour point is caused by the abundance of relatively short side chains, predominantly single methyl groups. To attain a lower pour point, one needs longer branches.⁶³ Together with the volatility values, we conclude that the oils could be used at temperatures above −10 °C. Both the heavy and light oils have VI values well above the typical mineral oil threshold, indicating their potential in niche applications requiring stable lubrication by suitable formulation to reduce their pour point and increase their thermal stability.

CONCLUSIONS

We demonstrated lubricant-range hydrocarbons from polypropylene plastic waste over Ru/TiO₂ catalysts at relatively low temperatures and modest reaction times with low gasification and a high oil yield (66–80% depending on the plastic). The catalyst is agnostic to the polypropylene nature (isotactic, amorphous, and different molecular weight), making similar products. However, the polymer microstructure and molecular weight strongly impact the kinetics of depolymerization. Excitingly, fundamental lubricant properties reveal that heavy

and light oils have potential as base oils for specific applications to enable a circular economy. Given the large lubricant market, the projected growth, and the high value of lubricants, this is an exciting upcycling strategy.

Ru/TiO₂ is a unique catalyst; it is commercially available and has a higher affinity for longer polymer chains. While known as a methane-producing catalyst, mild conditions enable profoundly high productivity to oil. Unlike the processive mechanism exposed for LDPE hydrogenolysis over Pt/SiO₂, PP over Ru/TiO₂ undergoes a dynamic adsorption/desorption and internal C–C bond breaking, lowering the molecular weight. Loss of stereoregularity of the initial melt appears as a prerequisite for hydrogenolysis. The polymer backbone demethylation leads to an oil reminiscent of an ethylene-propylene copolymer with reduced tertiary carbon and reactivity to cracking, a key aspect for minimal gas production. (De)hydrogenation in later stages is implicated in the rate-controlling chemistry. This dynamic reaction and product evolution appear responsible for the process selectivity to valuable oil over gas. The significant mechanistic and product differences found here from prior works expose exciting opportunities to engineer plastic recycling and upcycling.

MATERIALS AND METHODS

Catalyst Preparation. Catalysts were prepared using the wetness impregnation method. TiO₂ nanopowder (99.5%, US Research Nanomaterials, 5 nm, anatase) was first calcined at 450 °C for 4 h in air. Then TiO₂ was heated to 70 °C with the subsequent addition of ruthenium nitrosyl nitrate Ru(NO)(NO₃)₃ solution in diluted HNO₃ (Sigma-Aldrich no. 373567). The suspension was vigorously stirred with a glass rod for 1.5 h until complete dryness; then, it was stored in an oven at 100 °C for 12 h. Before the reaction, the catalyst was reduced in the flow of 50% H₂/He at 300 °C for 2 h (ramping rate 10 °C/min). The procedure on other supports, including SiO₂ (Davisil 646, Sigma-Aldrich) and CeO₂ (44960 Alfa Aesar), was the same. Boehmite (Pural-SB, Sasol) was calcined at 550 °C for 6 h in air to prepare the Al₂O₃ support. Ru/C (5% Ru loading, Sigma-Aldrich) was used as received.

Catalyst Characterization. The Ru loading was estimated using XRF analysis on a Rigaku Supermini 200 WDXRF in a He atmosphere. XRD patterns were obtained on a Bruker D8 diffractometer with 0.05° 2θ step size using Cu K_α radiation (λ 1.54 Å). N₂ sorption isotherms at −196 °C were recorded on Micromeritics ASAP 2020 instrument. Before measurements, the samples were degassed at 300 °C for 3 h. Photoelectron spectra were recorded on a Thermo Fisher K-Alpha+ machine with an Al K_α monochromatic source. Before measurements, the samples were reduced in 50% H₂/He flow at 300 °C and then deposited on a Cu foil. For binding energy reference, the C 1s line at 284.6 eV was used. TEM images were acquired on an Aberration Corrected Scanning/Transmission Electron Microscope, JEOL NEOARM TEM/STEM.

Feedstocks. Isotactic polypropylene (i-PP, average $M_w \sim 250\,000$, average $M_n \sim 67\,000$), low molecular weight isotactic polypropylene (l-PP, average $M_w \sim 12\,000$, average $M_n \sim 5000$), and amorphous PP (a-PP, 428175) were purchased from Sigma-Aldrich. 500 mL PP bottles were purchased from United Scientific (no. 33309). PP transparent bags (140 × 184 mm) were obtained from AIRSUNNY (item model number AS-57DZ-100).

Reaction Tests. An amount of freshly reduced catalyst was mechanically mixed with 2.0 g of polymer feedstock using a vortex mixer. The mixture was then transferred into a borosilicate liner of a 50 mL stainless-steel Parr reactor with a 0.7 mL stir bar. The mass ratio of polymer to catalyst was 20 or 40, corresponding to the catalyst loading of 100 or 50 mg. The Parr reactor was sealed and purged six times with pure H₂ at 50 bar, charged to 30 bar, and then heated to 250 °C (ramping rate 10°/min) using a band heater (Omega Eng.). Stirring was initiated after the temperature reached 160 ± 5 °C to first melt the polymer. The stirring speed was set at 300 rpm; additional experiments showed similar product yields and molecular weight distributions with 400 and 500 rpm but nearly no reactivity without stirring. Reactions were maintained for specified time intervals and then quickly quenched in an ice bath.

Product Analysis. After the temperature dropped below 10 °C, the gas from the reactor's headspace was transferred to a 1 L Tedlar gas sampling bag for analysis. Then the reactor was opened, liquid and solid residue were mixed with 20 mL of CH₂Cl₂, used as a solvent. This slurry was filtered (Whatman, 100 μm), and the solid residue was dried at room temperature overnight with complete evaporation of all CH₂Cl₂. The solvent was removed from the liquid fraction using a rotary evaporator. The solid and liquid fraction yields were quantified gravimetrically. A GC with an FID detector (Agilent 7890 Series, HP-volamine column) was used for gas analysis. Isotope analysis of gas products produced with D₂ as the reactant was performed on a Shimadzu GC-2010 GCMS instrument using Agilent PoraPLOT Q 10 m × 0.32 mm × 10 μm column. The concentration of hydrocarbons in the gas sample was calculated using a standard C₁–C₄ calibration mixture. The absolute amount of hydrocarbons in the gas was calculated using the ideal gas law.

A material balance was calculated as follows:

$$\text{carbon balance} = \frac{m_L + m_s + m_g}{m_{\text{initial}}} \times 100\%$$

where m_L , m_s , m_g , and m_{initial} are the mass of liquid, solid, gas, and initial polymer, respectively. The yield of the i -th group of products was calculated as

$$Y_i = \frac{m_i}{m_{\text{initial}}} \times 100\%$$

where m_i is the mass of the i -th group of products.

Liquid products were analyzed using gel permeation chromatography (GPC) using Styragel HR 4, HR 3, and HR 0.5 columns (dimensions 4.6 × 300 mm) in tandem using THF as solvents (0.3 mL/min flow rate) and a Waters 2414 refractive index detector (RID). The retention time was calibrated using Polystyrene Standards Kit (Waters, WAT058931).

ATR spectra of liquid and solid samples and the initial polymers were measured using a Nicolet Nexus 640 spectrometer with a Smart Orbit Diamond ATR accessory in the 4000–650 cm^{−1} range. Heteronuclear single quantum coherence (HSQC) nuclear magnetic resonance (NMR) spectra of liquid products were recorded at 25 °C on an Avance III 400 MHz NMR spectrometer (Bruker). Approximately 90 mg of the sample was mixed with 500 μL of toluene-*d*₈. Data processing was performed using the Mestrelab Research software (mNOVA).

The lubricant properties were evaluated according to the American Society for Testing and Materials (ASTM) methods. The kinematic viscosities at 100° and 40 °C (KV100 and KV40) were determined using the ASTM D445 method. The VI was calculated using the KV100 and KV40 following the ASTM D2270 method. The pour point (ASTM D5949), oxidation onset temperature (ASTM D6186) were measured at Southwest Research Institute (San Antonio, TX).

Catalyst Regeneration. For the reusability test, Ru/TiO₂ was first used in a regular run at 250 °C for 16 h with a polymer/catalyst ratio of 20 (results shown in Figure 2a). After the reaction, the catalyst and a small amount of solid residue were filtered and dried at room temperature overnight. Then, the powder was transferred to a tubular furnace and reduced in the flow of 50% H₂/He mixture at 300 °C for 2 h (ramp rate 10°/min).

■ ASSOCIATED CONTENT

■ Supporting Information

The Supporting Information is available free of charge at <https://pubs.acs.org/doi/10.1021/acscatal.1c00874>.

Details on catalyst screening, sample characterization results, such as XRD, HRTEM, and XPS, detailed GPC, ATR-IR, and NMR analysis, and isotope H/D analysis (PDF)

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Author Contributions

D.G.V. supervised the project. P.A.K. and S.L. conceived the idea and did preliminary experiments. P.A.K. designed the project and carried out most of the experiments. B.V. and C.W. carried out some experiments and contributed to the

discussions. J.D.S. measured and analyzed the NMR spectra. P.A.K. and D.G.V. wrote the manuscript. All authors discussed the results and assisted in the manuscript preparation.

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Notes

The authors declare the following competing financial interest(s): D.G.V., S.L., P.A.K., B.V., and C.W. have submitted patent applications. Other authors declare no competing interests.

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